Permanent-on-top holds for real matrices of rank two

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Abstract. Let A be a real symmetric positive semidefinite $n \times n$ matrix of rank at most two, and let $\Pi(A)$ be its Schur power matrix, the $n! \times n!$ matrix with entries $\Pi(A)_{\sigma,\tau} = \prod_i a_{\sigma(i),\tau(i)}$. We prove $0 \preceq \Pi(A) \preceq \operatorname{per}(A) I$ for every n: the permanent is the largest eigenvalue of the Schur power matrix. The permanent-on-top conjecture is known to fail for complex matrices of rank two and for real matrices of rank three, so this is the exact extent of its validity. The proof constructs, for each k, an explicit symmetric involution W on $\operatorname{Sym}^{n-2k}(\mathbb{R}^2) \otimes \operatorname{Sym}^n(\mathbb{R}^2)$ whose partial trace over the second factor is the identity, and uses it to bound every isotypic block of the permutation-averaged tensor operator by its fully symmetric block. Consequences on this matrix class include Lieb's permanental dominance inequality for every subgroup of S_n and every character, the Bapat–Sunder inequality $\operatorname{per}(A \circ B) \le \operatorname{per}(A) \prod_i b_{ii}$ for every complex positive semidefinite B, both conjectures on real rank-two matrices in Drury's 2018 paper, and the permanental compound conjectures of Bapat–Sunder and Pate.

1. Introduction

Let $A = (a_{ij})$ be an $n \times n$ complex Hermitian positive semidefinite matrix and let $\operatorname{per}(A) = \sum_{\sigma \in S_n} \prod_{i=1}^n a_{i,\sigma(i)}$ be its permanent. The *Schur power matrix* of A is the $n! \times n!$ matrix

$$\Pi(A)_{\sigma,\tau} = \prod_{i=1}^{n} a_{\sigma(i),\tau(i)}, \quad \sigma, \tau \in S_n. \tag{1}$$

Schur proved in 1918 that $\Pi(A)$ is positive semidefinite, that $\operatorname{per}(A)$ is one of its eigenvalues, with the all-ones vector as eigenvector, and that $\det(A)$ is its smallest eigenvalue [1]. The *permanent-on-top conjecture* of Soules [2], [3] asserted that $\operatorname{per}(A)$ is also the largest eigenvalue:

$$\Pi(A) \preceq \operatorname{per}(A) I. \tag{2}$$

A proof would have implied Lieb's permanental dominance conjecture [4] and a whole family of permanent inequalities; see Wanless [5] for the implications. The conjecture holds for $n \le 3$ [6]. It is false in general: Shchesnovich [7] found a complex counterexample of order five and rank two, Drury [8] found a real counterexample of order sixteen and rank three, and Tran [9] found a complex counterexample of order five and rank two that can be checked by hand.

This paper proves that permanent-on-top is exactly true on the remaining class.

Theorem 1.1. *Let A be a real symmetric positive semidefinite $n \times n$ matrix of rank at most two. Then*

$$0 \preceq \Pi(A) \preceq \operatorname{per}(A) I_{n!}. \tag{3}$$

In particular $\operatorname{per}(A)$ *is the largest eigenvalue of* $\Pi(A)$.

Together with the counterexamples of Drury and Tran, Theorem 1.1 gives a complete picture in terms of rank and field:

class	permanent-on-top	reference
real, rank ≤ 2	holds for all n	Theorem 1.1
real, rank 3	fails ($n = 16$)	Drury [8]
complex, rank 2	fails ($n = 5$)	Shchesnovich [7], Tran [9]

Every implication in Wanless's diagram now applies on the real rank-two class. We record the consequences in Section 7. The most notable is that Lieb's conjecture holds there in full generality: for every subgroup $H \le S_n$ and every character χ of H,

$$0 \le \frac{1}{\chi(1)} \sum_{h \in H} \chi(h) \prod_{i} a_{i,h(i)} \le \operatorname{per}(A). \tag{4}$$

Further, the Hadamard product inequality $\operatorname{per}(A \circ B) \le \operatorname{per}(A) \prod_i b_{ii}$ holds with the second factor an arbitrary complex positive semidefinite matrix of the same order. Drury [10] stated as Conjecture 2 that $\operatorname{per}(A \circ A) \le \operatorname{per}(A)$ for real rank-two correlation matrices, and conjectured (as recorded in [5]) that the Bapat–Sunder Hadamard inequality holds for real rank-two matrices. Both follow. So do the compound conjectures of Bapat–Sunder [6] and Pate [11] on this class, including the cofactor bound $\left[a_{ij} \operatorname{per} A(i|j)\right] \preceq \operatorname{per}(A) I$.

The method. Write A as the Gram matrix of vectors $v_1, \dots, v_n \in \mathbb{R}^2$ and let $\psi = v_1 \otimes \cdots \otimes v_n \in (\mathbb{R}^2)^{\otimes n}$. Then $\Pi(A)$ is the Gram matrix of the $n!$ vectors $U_\sigma \psi$, where U_σ permutes tensor factors. Its nonzero spectrum coincides with that of the $2^n \times 2^n$ operator $n! Q$, $Q = (n!)^{-1} \sum_\sigma |U_\sigma \psi\rangle\langle U_\sigma \psi|$. Under $GL_2(\mathbb{R})$ the space $(\mathbb{R}^2)^{\otimes n}$ decomposes into isotypic blocks indexed by $k = 0, \dots, \lfloor n/2 \rfloor$, the k-th block being $\det^k \otimes \operatorname{Sym}^{n-2k}$ with multiplicity. A polarization identity writes Q as a signed sum of tensor powers of real positive semidefinite 2×2 matrices, so Q is block diagonal with blocks R_k. The block R_0 on the fully symmetric tensors is rank one with trace $\operatorname{per}(A)/n!$. The heart of the proof is an explicit linear map $\Phi_{n,k}$ carrying R_0 to R_k and satisfying $\Phi_{n,k}(S) \preceq (\operatorname{tr} S) I$ for $S \succeq 0$. It is built from a symmetric involution $W_{n,k}$ on $\operatorname{Sym}^{n-2k} \otimes \operatorname{Sym}^n$ whose partial trace over the second factor is the identity. The construction of $W_{n,k}$ and the verification of its three properties occupy Sections 3 and 4.

The real hypothesis enters at exactly one point: the covariance identity $\Phi_{n,k}(\rho_n(M)) = (\det M)^k \rho_{n-2k}(M)$ is proved for $M = g g^T$ with g real, and Hermitian rank-one factors $v v^*$ are not of this form. This is where Tran's complex example escapes.

Concurrently with this work, Kunisky, Spielman and Yu [12] sharpened Bang's inequality for real rank-two matrices using the same binary Clebsch–Gordan decomposition. Their inequality is different from (3) and is not used here.

2. Preliminaries

Throughout, $E = \mathbb{R}^2$ with the standard inner product $x \cdot y$, and $[x, y] = x_1 y_2 - x_2 y_1$. For $d \ge 0$ let $V_d = \operatorname{Sym}^d(E) \subset E^{\otimes d}$ with the inner product inherited from the tensor product, $V_0 = \mathbb{R}$. For $u \in E$ write $u^d = u^{\otimes d} \in V_d$. These *coherent vectors* span V_d and satisfy $\langle u^d, v^d \rangle = (u \cdot v)^d$. Every $g \in GL_2(\mathbb{R})$ acts on V_d by $\rho_d(g) = g^{\otimes d}|_{V_d}$; thus $\rho_d(g) u^d = (gu)^d$ and $\rho_d(g)^T = \rho_d(g^T)$.

Let e_1, e_2 be the standard basis of E. For $0 \le a \le d$ let $e_a^{(d)}$ be the unit vector proportional to the symmetrization of $e_1^{\otimes a} \otimes e_2^{\otimes (d-a)}$. These form an orthonormal basis of V_d, and

$$u^d = \sum_{a=0}^{d} \sqrt{\binom{d}{a}}\, u_1^a u_2^{d-a}\, e_a^{(d)}. \tag{5}$$

The $\mathfrak{sl}_2$ operators. On V_d define

$$H_d e_a^{(d)} = \left(a - \frac{d}{2}\right) e_a^{(d)}, \quad E_d e_a^{(d)} = \sqrt{(a+1)(d-a)}\, e_{a+1}^{(d)}, \quad F_d = E_d^T. \tag{6}$$

These satisfy $[H_d, E_d] = E_d$, $[H_d, F_d] = -F_d$, $[E_d, F_d] = 2H_d$, and they are the derivatives of ρ_d: writing X_{ij} for the matrix units,

$$\mathrm{d}\rho_d(X_{12}) = E_d, \quad \mathrm{d}\rho_d(X_{21}) = F_d, \quad \mathrm{d}\rho_d(X_{11} - X_{22}) = 2H_d. \tag{7}$$

Indeed the derivative of $g^{\otimes d}$ at the identity acts on a symmetrized word by acting on each factor in turn; on the word with a letters e_1, the matrix unit X_{12} turns each of the $d-a$ letters e_2 into e_1, producing $(a+1)$ times the symmetrized word with $a+1$ letters e_1, and the normalizations in (5) convert this count into $\sqrt{(a+1)(d-a)}$.

Clebsch–Gordan. The following decomposition is classical; see [13, Section 11.2]. We include the short argument because the real orthogonality of the blocks is what we use.

Lemma 2.1. *As an orthogonal direct sum of $GL_2(\mathbb{R})$-modules,*

$$E^{\otimes n} \cong \bigoplus_{k=0}^{\lfloor n/2 \rfloor} \left(\det{}^{\otimes k} \otimes V_{n-2k}\right) \otimes \mathbb{R}^{d_{n,k}}, \quad d_{n,k} = \binom{n}{k} - \binom{n}{k-1}. \tag{8}$$

Consequently, for every real symmetric positive semidefinite M,

$$M^{\otimes n} \cong \bigoplus_{k} (\det M)^k \rho_{n-2k}(M) \otimes I_{d_{n,k}}. \tag{9}$$

Proof. Let H, E, F act on $E^{\otimes n}$ as the sums over tensor positions of H_1, E_1, F_1. They satisfy the $\mathfrak{sl}_2$ relations, H is symmetric, and $F = E^T$. Weight spaces of H are spanned by words with a prescribed number of letters e_1, so the weight h has multiplicity $\binom{n}{n/2+h}$. A highest-weight vector w of weight j generates by $w, Fw, F^2w, \dots$ an F-stable subspace on which the relations force $EF^r w = r(2j - r + 1)F^{r-1}w$; this subspace is E-stable, has dimension $2j+1$, and its normalized basis realizes (6) for $d = 2j$. The orthogonal complement of an (H, E, F)-stable subspace is stable because H is symmetric and E, F are mutually adjoint, so induction decomposes $E^{\otimes n}$ orthogonally into copies of the V_d with $d \equiv n \bmod 2$. Comparing the multiplicities of consecutive weights gives $\binom{n}{k} - \binom{n}{k-1}$ copies of V_{n-2k}. For $g \in GL_2(\mathbb{R})$ with $\det g > 0$ write $g = (\det g)^{1/2} \exp(X)$ with $\operatorname{tr} X = 0$; the $\mathfrak{sl}_2$ action identifies $\exp(X)^{\otimes n}$ with $\rho_{n-2k}(\exp X)$ on each copy of V_{n-2k}, and the scalar contributes $(\det g)^{n/2} = (\det g)^k (\det g)^{(n-2k)/2}$, which is $(\det g)^k \rho_{n-2k}\big((\det g)^{1/2} I\big)$. This proves (8) and (9) for positive definite M; the identity is polynomial in M, so it persists for singular M. $\square$

3. The involution

Fix $n \geq 0$ and $0 \leq k \leq \lfloor n/2 \rfloor$, and put $m = n - 2k$. Define a real operator $W = W_{n,k}$ on $V_m \otimes V_n$ through its values on coherent vectors:

$$\langle u^m \otimes x^n, W(v^m \otimes y^n)\rangle = \sum_{\ell=0}^{m} c_\ell\, [x,y]^{2k+\ell} [u,v]^{\ell} (x \cdot u)^{m-\ell} (y \cdot v)^{m-\ell}, \tag{10}$$

$$c_\ell = (-1)^{k+\ell} \binom{n}{k} \binom{m}{\ell} \binom{2k+\ell}{k}^{-1}. \tag{11}$$

The right side of (10) is a polynomial in (u, x, v, y), homogeneous of degrees (m, n, m, n): each term has degree $2k + \ell + (m - \ell) = n$ in x and in y, and degree $\ell + (m - \ell) = m$ in u and in v. Since $\langle u^m \otimes x^n, \cdot \rangle$ identifies such polynomials with linear functionals on $V_m \otimes V_n$ and the coherent vectors span, (10) determines W uniquely.

Theorem 3.1. *W is symmetric, $W^2 = I$, and its partial trace over V_n is the identity of V_m:*

$$W^T = W, \quad W^2 = I, \quad \mathrm{Tr}_{V_n} W = I_{V_m}. \tag{12}$$

In particular $-I \preceq W \preceq I$.

The proof occupies the rest of this section and the next. For $k = 0$, the bracket identity

$$[x, y][u, v] = (x \cdot u)(y \cdot v) - (x \cdot v)(y \cdot u) \tag{13}$$

reduces (10) to $(x \cdot v)^n (y \cdot u)^n$, so $W_{n,0}$ is the swap of the two factors of $V_n \otimes V_n$, and (12) is immediate. The arguments below apply to all k, but the partial-trace computation of Section 4 is needed only for $k \geq 1$.

3.1. Symmetry and covariance

Interchanging (u, x) with (v, y) in (10) multiplies each bracket factor by $(-1)^{2k+\ell}(-1)^{\ell} = 1$ and interchanges the two dot-product factors, so the kernel is symmetric and $W^T = W$.

For $g \in GL_2(\mathbb{R})$ put $B = \rho_m(g)$, $C = \rho_n(g)$ and $\mathcal{R}(g) = B^{-T} \otimes C$. Replacing u, v by $g^{-T}u, g^{-T}v$ and x, y by gx, gy leaves every dot product in (10) unchanged, multiplies $[x, y]$ by $\det g$ and $[u, v]$ by $(\det g)^{-1}$, hence multiplies each term by $(\det g)^{2k}$. Since $\left(g^{-T}u\right)^m = B^{-T}u^m$ and $(gx)^n = Cx^n$,

$$\mathcal{R}(g)^T W \mathcal{R}(g) = (\det g)^{2k} W. \tag{14}$$

On $V_m \otimes V_n$ define

$$\boldsymbol{H} = I \otimes H_n - H_m \otimes I, \quad \boldsymbol{E} = I \otimes E_n - F_m \otimes I, \quad \boldsymbol{F} = \boldsymbol{E}^T = I \otimes F_n - E_m \otimes I. \tag{15}$$

These satisfy the $\mathfrak{sl}_2$ relations, being the action on $V_m^* \otimes V_n$ with V_m^* identified with V_m by the inner product. Differentiating (14) along $g = \exp(tX)$ with $\mathrm{tr}\, X = 0$, using (7) and $\mathrm{d}\rho_m(X)^T = \mathrm{d}\rho_m(X^T)$, gives $\mathfrak{X}^T W + W\mathfrak{X} = 0$ for $\mathfrak{X} = I \otimes \mathrm{d}\rho_n(X) - \mathrm{d}\rho_m(X)^T \otimes I$. The three choices $X = (X_{11} - X_{22})/2$, X_{12}, X_{21} give $\mathfrak{X} = \boldsymbol{H}, \boldsymbol{E}, \boldsymbol{F}$ with transposes $\boldsymbol{H}, \boldsymbol{F}, \boldsymbol{E}$, hence

$$W\boldsymbol{H} = -\boldsymbol{H}W, \quad W\boldsymbol{E} = -\boldsymbol{F}W, \quad W\boldsymbol{F} = -\boldsymbol{E}W. \tag{16}$$

So W maps the $\boldsymbol{H}$-eigenspace of weight h onto that of weight $-h$.

3.2. The zero-weight space

The eigenvalues of $\boldsymbol{H}$ on $e_a^{(m)} \otimes e_c^{(n)}$ are $(c - n/2) - (a - m/2) = c - a - k$, so all weights are integers and the zero-weight space has the orthonormal basis

$$b_a = e_a^{(m)} \otimes e_{a+k}^{(n)}, \quad 0 \leq a \leq m. \tag{17}$$

Lemma 3.2. $Wb_0 = b_0$.

Proof. By (16), Wb_0 lies in the zero-weight space, so it suffices to compute $\langle b_0, Wb_b \rangle$ for $0 \leq b \leq m$. Take $u = e_2$ in (10), so that $u^m = e_0^{(m)}$, $x \cdot u = x_2$ and $[u, v] = -v_1$; then extract the coefficient of $x_1^k x_2^{n-k}$, of $v_1^b v_2^{m-b}$ and of $y_1^{b+k} y_2^{n-b-k}$, which by (5) equals $\sqrt{\binom{n}{k}\binom{m}{b}\binom{n}{b+k}}\, \langle b_0, Wb_b \rangle$. In the term of index ℓ, the factor x_1^k must come from $[x, y]^{2k+\ell} = (x_1 y_2 - x_2 y_1)^{2k+\ell}$, contribut-

ing $\binom{2k+\ell}{k} x_1^k y_2^k (-x_2 y_1)^{k+\ell}$; the factor $(-v_1)^\ell (y\cdot v)^{m-\ell}$ then contributes v_1^b exactly through $\binom{m-\ell}{b-\ell}(y_1 v_1)^{b-\ell}(y_2 v_2)^{m-b}$. The exponents of x_2, y_1, y_2 come out as $n-k$, $b+k$, $n-b-k$ automatically. Hence the coefficient is

$$(-1)^k \sum_{\ell=0}^{b} c_\ell \binom{2k+\ell}{k}\binom{m-\ell}{b-\ell} = \binom{n}{k}\sum_{\ell=0}^{b}(-1)^\ell \binom{m}{\ell}\binom{m-\ell}{b-\ell} = \binom{n}{k}\binom{m}{b}(1-1)^b,$$

using (11) and $\binom{m}{\ell}\binom{m-\ell}{b-\ell} = \binom{m}{b}\binom{b}{\ell}$. For $b=0$ this equals $\binom{n}{k}$, and division by the normalization $\binom{n}{k}$ gives $\langle b_0, W b_0\rangle = 1$; for $b \geq 1$ it vanishes. By symmetry of W, $W b_0 = \sum_b \langle b_b, W b_0\rangle b_b = b_0$.□

Lemma 3.3. *W fixes every vector of the zero-weight space.*

Proof. Let $\boldsymbol{C} = \boldsymbol{H}^2 + \frac{1}{2}(\boldsymbol{EF}+\boldsymbol{FE})$ be the Casimir operator. From (16), $W\boldsymbol{H}^2 = \boldsymbol{H}^2 W$ and $W(\boldsymbol{EF}+\boldsymbol{FE}) = (\boldsymbol{FE}+\boldsymbol{EF})W$, so W commutes with $\boldsymbol{C}$. Since $\boldsymbol{C}$ commutes with $\boldsymbol{H}$ it preserves the zero-weight space, where $\boldsymbol{H}^2 = 0$. Computing $\boldsymbol{E}b_a$ and $\boldsymbol{F}b_a$ from (6) and (15), the matrix of $\boldsymbol{C}$ in the basis (17) is tridiagonal with off-diagonal entries

$$\langle b_{a+1}, \boldsymbol{C} b_a\rangle = -\sqrt{(a+1)(m-a)(a+k+1)(n-a-k)}, \quad 0 \leq a < m, \tag{18}$$

all nonzero because $n-a-k \geq k+1$. Hence $\boldsymbol{C}^j b_0$ has a nonzero component along b_j and no component along b_i for $i>j$, so $b_0, \boldsymbol{C}b_0, \ldots, \boldsymbol{C}^m b_0$ span the zero-weight space. Each is fixed by W, by Lemma 3.2 and $W\boldsymbol{C} = \boldsymbol{C}W$. □

3.3. The involution identity

Lemma 3.4. $W^2 = I$.

Proof. By (16), W^2 commutes with $\boldsymbol{H}$, $\boldsymbol{E}$ and $\boldsymbol{F}$, and by Lemma 3.3 it is the identity on the zero-weight space. If y has weight $h>0$ and $\boldsymbol{F}y = 0$, then $0 = \langle y, \boldsymbol{EF}y\rangle = \langle y, (\boldsymbol{FE}+2\boldsymbol{H})y\rangle = \|\boldsymbol{E}y\|^2 + 2h\,\|y\|^2$, so $y=0$; thus $\boldsymbol{F}$ is injective on positive weights and its adjoint $\boldsymbol{E}$ maps the weight-$(h-1)$ space onto the weight-h space for every $h \geq 1$. Symmetrically $\boldsymbol{F}$ maps weight $-(h-1)$ onto weight $-h$. Since all weights are integers, every vector is a sum of images of zero-weight vectors under powers of $\boldsymbol{E}$ or of $\boldsymbol{F}$, and W^2 is the identity on all of them. □

Symmetry and $W^2 = I$ give $-I \preceq W \preceq I$. It remains to prove the partial-trace identity.

4. The partial trace

Assume $k \geq 1$. Two tools are needed: a Beta-integral form of the kernel, and Gaussian moments.

Beta form of the kernel. Since $\binom{2k+\ell}{k}^{-1} = k\int_0^1 t^{k+\ell}(1-t)^{k-1}\,\mathrm{d}t$, and since (13) gives $[x,y]^\ell [u,v]^\ell (x\cdot u)^{m-\ell}(y\cdot v)^{m-\ell} = (a-b)^\ell a^{m-\ell}$ with $a = (x\cdot u)(y\cdot v)$ and $b = (x\cdot v)(y\cdot u)$, the binomial theorem turns (10) into

$$K(u,x;v,y) := \langle u^m \otimes x^n, W(v^m \otimes y^n)\rangle,$$

$$K(u,x;v,y) = (-1)^k \binom{n}{k} k\,[x,y]^{2k}\int_0^1 t^k(1-t)^{k-1}((1-t)a+tb)^m\,\mathrm{d}t. \tag{19}$$

Gaussian resolution of the identity. Let $Z = (Z_1, Z_2)$ be a standard complex Gaussian vector, $\mathbb{E}\,|Z_i|^2 = 1$. By (5) the components of Z^n in the basis $e_a^{(n)}$ are $\sqrt{\binom{n}{a}} Z_1^a Z_2^{n-a}$, and $\mathbb{E}\big[\overline{Z_1}^a \overline{Z_2}^{n-a} Z_1^b Z_2^{n-b}\big] = \delta_{ab}\, a!(n-a)!$. Hence $\mathbb{E}\big[\overline{Z}^n \otimes Z^n\big] = n!\sum_a e_a^{(n)} \otimes e_a^{(n)}$, and extending K bilinearly to complex arguments,

$$\left\langle u^m, \left(\operatorname{Tr}_{V_n} W\right) v^m \right\rangle = \frac{1}{n!} \mathbb{E}\, K\big(u, \overline{Z}; v, Z\big). \tag{20}$$

Put $\alpha = u \cdot v$, $\beta = [u, v]$, $J = \left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$ and $M_t = (1-t)uv^T + tvu^T$. With $x = \overline{Z}$, $y = Z$ one has $[x, y] = Z^* J Z$ and $(1-t)a + tb = Z^* M_t Z$. For a complex 2×2 matrix N of small norm, $\mathbb{E} \exp(Z^* N Z) = \det(I - N)^{-1}$, a Gaussian integral; both sides are analytic in the entries of N, so the identity holds coefficientwise as formal power series. A direct computation of the determinant gives

$$\mathbb{E} \exp(s Z^* J Z + r Z^* M_t Z) = \frac{1}{1 - r\alpha + s^2 + (1-2t) sr\beta - t(1-t) r^2 \beta^2}. \tag{21}$$

Indeed, with $M_t = \big(m_{ij}\big)$ one has $m_{11} + m_{22} = \alpha$, $m_{12} - m_{21} = (1-2t)\beta$ and $m_{11} m_{22} - m_{12} m_{21} = -t(1-t)\beta^2$, and the determinant of $I - sJ - rM_t$ expands accordingly. Note that the denominator of (21) factors as $a_0 + (s + (1-t)v_0)(s - t v_0)$ with $a_0 = 1 - r\alpha$ and $v_0 = r\beta$.

Lemma 4.1. *For $k \geq 1$, as formal power series in s and v_0 with a_0 invertible,*

$$\big[s^{2k}\big] \int_0^1 \frac{t^k (1-t)^{k-1}}{a_0 + (s + (1-t)v_0)(s - t v_0)} \, \mathrm{d}t = (-1)^k \frac{k!(k-1)!}{(2k)!} a_0^{-k-1}. \tag{22}$$

In particular the coefficient does not depend on v_0.

Proof. Expand $(a_0 + w)^{-1} = \sum_p (-1)^p w^p a_0^{-p-1}$ with $w = (s + (1-t)v_0)(s - t v_0)$, homogeneous of degree two in (s, v_0). The coefficient of $s^{2k} v_0^{2l}$ can only come from $p = k + l$, where

$$\big[s^{2k} v_0^{2l}\big]\, w^{k+l} = \sum_i \binom{k+l}{i} \binom{k+l}{2k-i} (1-t)^{k+l-i} (-t)^{l-k+i}.$$

Integrating against $t^k (1-t)^{k-1}$ gives, up to the factor $(-1)^{k+l} a_0^{-k-l-1}$,

$$\sum_i (-1)^{l-k+i} \binom{k+l}{i} \binom{k+l}{2k-i} \frac{(l+i)!(2k+l-i-1)!}{(2k+2l)!}, \tag{23}$$

the sum running over the i for which all factorials are of nonnegative integers. For $l = 0$ only $i = k$ contributes, giving $k!(k-1)!/(2k)!$, which is the right side of (22). For $l \geq 1$ write

$$\binom{k+l}{i} \binom{k+l}{2k-i} (l+i)!(2k+l-i-1)! = \frac{(k+l)!^2}{(2k)!} \binom{2k}{i} P_l(i), \quad P_l(i) = \prod_{a=1}^{k} (l - k + i + a) \prod_{b=1}^{k-1} (k + l - i + b).$$

So (23) is a constant multiple of $\sum_i (-1)^i \binom{2k}{i} P_l(i)$. The range of i may be extended to all of $0 \leq i \leq 2k$: for $i < k - l$ the first product contains the factor 0, and for $i > k + l$ the second does. But P_l is a polynomial of degree $2k - 1$, and the $2k$-th finite difference of such a polynomial vanishes. Hence every coefficient with $l \geq 1$ is zero. $\square$

Proof. (of the partial-trace identity in Theorem 3.1.) By (20), (19) and (21),

$$\left\langle u^m, \left(\operatorname{Tr}_{V_n} W\right) v^m \right\rangle = \frac{(-1)^k \binom{n}{k} k(2k)! m!}{n!} \big[s^{2k} r^m\big] \int_0^1 \frac{t^k (1-t)^{k-1}}{1 - r\alpha + s^2 + (1-2t) sr\beta - t(1-t) r^2 \beta^2} \, \mathrm{d}t,$$

since $\mathbb{E}\big[(Z^* J Z)^{2k} (Z^* M_t Z)^m\big] = (2k)! m! \big[s^{2k} r^m\big]$ of the left side of (21). By Lemma 4.1 with $a_0 = 1 - r\alpha$ and $v_0 = r\beta$, the coefficient of s^{2k} is $(-1)^k k!(k-1)!(2k)!^{-1} (1 - r\alpha)^{-k-1}$, whose coefficient of r^m is $\binom{k+m}{m} \alpha^m$. Therefore

$$\left\langle u^m, \left(\operatorname{Tr}_{V_n} W\right) v^m \right\rangle = \frac{\binom{n}{k}(k!)^2 m! \binom{k+m}{m}}{n!} \alpha^m = \alpha^m = \langle u^m, v^m \rangle,$$

because $\binom{n}{k}(k!)^2 m!\binom{k+m}{m} = n!\, k!(k+m)!/(n-k)! = n!$ when $n-k = m+k$. Coherent vectors span V_m, so $\operatorname{Tr}_{V_n} W = I_{V_m}$. □

This completes the proof of Theorem 3.1.

5. A trace-dominating map between symmetric powers

For a real symmetric operator S on V_n define

$$\Phi_{n,k}(S) = \operatorname{Tr}_{V_n}\big(W_{n,k}(I \otimes S)\big), \tag{24}$$

an operator on V_m, $m = n - 2k$. Partial trace over the second factor is cyclic with respect to operators of the form $I \otimes S$, so $\Phi_{n,k}(S)$ is symmetric.

Lemma 5.1. *For every real symmetric $S \succeq 0$ on V_n,*

$$\Phi_{n,k}(S) \preceq (\operatorname{tr} S)\, I_{V_m}. \tag{25}$$

Proof. For $z \in V_m$, $z^T \Phi(S) z = \operatorname{Tr}\big(W(zz^T \otimes S)\big)$. The operator $zz^T \otimes S$ is positive semidefinite and $W \preceq I$ by Theorem 3.1, so $\operatorname{Tr}\big(W(zz^T \otimes S)\big) \le \operatorname{Tr}\big(zz^T \otimes S\big) = \|z\|^2 \operatorname{tr} S$. □

Nothing is claimed about Φ preserving positivity, and nothing of the kind is needed.

Lemma 5.2. *For every real symmetric positive semidefinite 2×2 matrix M,*

$$\Phi_{n,k}(\rho_n(M)) = (\det M)^k \rho_m(M). \tag{26}$$

Proof. Let $g \in GL_2(\mathbb{R})$, $B = \rho_m(g)$, $C = \rho_n(g)$. Multiplying (14) on the left by $B \otimes I$ and on the right by $B^T \otimes I$ gives $(I \otimes C^T) W (I \otimes C) = (\det g)^{2k} (B \otimes I) W (B^T \otimes I)$. Taking the partial trace against $I \otimes S$ and using cyclicity in the traced factor,

$$\Phi(CSC^T) = (\det g)^{2k} B\, \Phi(S)\, B^T.$$

Put $S = I$. Then $\Phi(I) = \operatorname{Tr}_{V_n} W = I$ by Theorem 3.1, and $CC^T = \rho_n(gg^T)$, $BB^T = \rho_m(gg^T)$, so $\Phi(\rho_n(gg^T)) = (\det g)^{2k} \rho_m(gg^T)$. Every positive definite M is gg^T with g real and $(\det g)^{2k} = (\det M)^k$; singular M follow by continuity. □

Lemma 5.2 is the only place where the real hypothesis is used, and it is used essentially: the factorization $M = gg^T$ with g real is what turns the transpose in (14) into the matrix M itself.

6. Proof of Theorem 1.1

Let $A = \big(v_i \cdot v_j\big)$ be the Gram matrix of $v_1, \dots, v_n \in \mathbb{R}^2$; every real symmetric positive semidefinite matrix of rank at most two is of this form. Put $\psi = v_1 \otimes \cdots \otimes v_n \in E^{\otimes n}$, $D_i = v_i v_i^T$, and let U_σ be the operator permuting the tensor factors of $E^{\otimes n}$ according to σ. Define

$$Q = \frac{1}{n!} \sum_{\sigma \in S_n} |U_\sigma \psi\rangle\langle U_\sigma \psi| \ . \tag{27}$$

Polarization. Expanding $\big(\sum_{i \in J} D_i\big)^{\otimes n}$ into words and applying inclusion–exclusion over $J \subseteq [n]$ cancels every word in which some D_i is missing, leaving exactly the $n!$ words using each D_i once. Since $|U_\sigma \psi\rangle\langle U_\sigma \psi| = D_{\sigma(1)} \otimes \cdots \otimes D_{\sigma(n)}$,

$$Q = \frac{1}{n!} \sum_{J \subseteq [n]} (-1)^{n-|J|} \left(\sum_{i \in J} D_i \right)^{\otimes n}. \tag{28}$$

Each $M_J = \sum_{i \in J} D_i$ is real symmetric positive semidefinite.

Block structure. By Lemma 2.1 applied to each term of (28), Q is block diagonal for the decomposition (8), with blocks $R_k \otimes I_{d_{n,k}}$ where

$$R_k = \frac{1}{n!} \sum_{J} (-1)^{n-|J|} (\det M_J)^k \rho_{n-2k}(M_J). \tag{29}$$

The $k = 0$ block lives on V_n, which occurs with multiplicity one. Let $S = R_0$. Applying Lemma 5.2 to each term of (29) and using linearity of $\Phi_{n,k}$,

$$R_k = \Phi_{n,k}(S) \quad (0 \le k \le \lfloor n/2 \rfloor). \tag{30}$$

The symmetric block. Let P be the orthogonal projection of $E^{\otimes n}$ onto V_n. Since $PU_\sigma = P$, compressing (27) gives $S = PQP = |P\psi\rangle\langle P\psi|$, a rank-one positive semidefinite operator with

$$\operatorname{tr} S = \langle \psi, P\psi \rangle = \frac{1}{n!} \sum_{\sigma} \langle \psi, U_\sigma \psi \rangle = \frac{1}{n!} \sum_{\sigma} \prod_{i} v_i \cdot v_{\sigma(i)} = \frac{\operatorname{per}(A)}{n!}. \tag{31}$$

Conclusion. Lemma 5.1, (30) and (31) give $R_k \preceq (\operatorname{per}(A)/n!) I$ for every k, hence

$$0 \preceq Q \preceq \frac{\operatorname{per}(A)}{n!} I. \tag{32}$$

Let T be the $2^n \times n!$ matrix with columns $U_\sigma \psi$. Then $TT^T = n! Q$ and $T^T T$ has entries $\langle U_\sigma \psi, U_\tau \psi \rangle = \prod_i a_{\sigma(i),\tau(i)}$, that is, $T^T T = \Pi(A)$. The two products have the same nonzero eigenvalues, so (32) gives $0 \preceq \Pi(A) \preceq \operatorname{per}(A) I$. Since the all-ones vector is an eigenvector of $\Pi(A)$ with eigenvalue $\operatorname{per}(A)$, the permanent is the largest eigenvalue. $\square$

7. Consequences

Throughout this section A is real symmetric positive semidefinite of rank at most two and of order n.

Corollary 7.1 (Lieb's inequality on the class). *For every subgroup $H \le S_n$ and every character χ of H, irreducible or not, real-valued or not,*

$$0 \le \frac{1}{\chi(1)} \sum_{h \in H} \chi(h) \prod_{i=1}^{n} a_{i,h(i)} \le \operatorname{per}(A). \tag{33}$$

Proof. The matrix $\Pi_H(A) = \big(\Pi(A)_{\sigma,\tau}\big)_{\sigma,\tau \in H}$ is a principal submatrix of $\Pi(A)$, so $0 \preceq \Pi_H(A) \preceq \operatorname{per}(A) I$. Choose a unitary representation with character χ; then $K_{\sigma,\tau} = \chi(\sigma^{-1}\tau)$ is the Gram matrix of the representing matrices in the Hilbert–Schmidt inner product, hence positive semidefinite, with trace $|H|\,\chi(1)$. Writing $m_h(A) = \prod_i a_{i,h(i)}$ one has $\Pi(A)_{\sigma,\tau} = m_{\tau\sigma^{-1}}(A)$, and since χ is a class function and A is symmetric, $\operatorname{Tr}(\Pi_H(A) K) = |H| \sum_{h \in H} \chi(h) m_h(A)$. The inequality $0 \le \operatorname{Tr}(\Pi_H(A) K) \le \operatorname{per}(A) \operatorname{Tr} K$ is (33). $\square$

Corollary 7.2 (Hadamard products). *For every complex Hermitian positive semidefinite B of order n,*

$$\operatorname{per}(A \circ B) \le \operatorname{per}(A) \prod_{i=1}^{n} b_{ii}. \tag{34}$$

In particular $\operatorname{per}(A \circ A) \le \operatorname{per}(A)$ *when* A *is a correlation matrix, which is Conjecture 2 of Drury [10].*

Proof. $\Pi(B)$ is positive semidefinite, hence so is $\Pi(B)^{T}$. Now $\operatorname{Tr}(\Pi(A)\Pi(B)^{T}) = \sum_{\sigma,\tau} \prod_i a_{\sigma(i)\tau(i)} b_{\sigma(i)\tau(i)} = n!\operatorname{per}(A \circ B)$ and $\operatorname{Tr}(\Pi(B)^{T}) = n! \prod_i b_{ii}$. Apply $0 \preceq \Pi(A) \preceq \operatorname{per}(A) I$. $\square$

The second factor in (34) is unrestricted in rank and need not be real. Drury's conjecture, as recorded by Wanless [5], that the Bapat–Sunder inequality $\operatorname{per}(A \circ B) \le \operatorname{per}(A) \prod_i b_{ii}$ holds for real rank-two A, is thus proved in a stronger form.

Corollary 7.3 (Permanental compounds). *For* $1 \le s \le n$ *let* $\mathcal{C}_s(A)$ *be the* $\binom{n}{s} \times \binom{n}{s}$ *matrix with entries* $\operatorname{per} A[I,J] \operatorname{per} A[I^{c}, J^{c}]$ *over* s*-subsets* I, J*. Then* $0 \preceq \mathcal{C}_s(A) \preceq \operatorname{per}(A) I$*. In particular the cofactor matrix* $\left[a_{ij} \operatorname{per} A(i|j)\right]$ *has largest eigenvalue* $\operatorname{per}(A)$*.*

Proof. Let $\mathcal{J}$ map a function f on s-subsets to the function $\sigma \mapsto f(\sigma(\{1,\dots,s\}))$ on S_n. Summing (1) over the permutations with a prescribed image of $\{1,\dots,s\}$ gives $\Pi(A)\mathcal{J} = \mathcal{J}\mathcal{C}_s(A)$, and $\mathcal{J}^{T}\mathcal{J} = s!(n-s)!\, I$. Hence $\mathcal{C}_s(A) = (s!(n-s)!)^{-1}\mathcal{J}^{T}\Pi(A)\mathcal{J}$ is a compression of $\Pi(A)$ by an isometry. $\square$

Corollary 7.3 is Conjecture 10 of Pate [11] and, for $s = 1$, Conjecture 3 of Bapat and Sunder [6], on the real rank-two class. Both are false in general [9], [10], [14].

8. Sharpness

The hypotheses of Theorem 1.1 cannot be relaxed in either direction.

Real rank three. Drury [8] exhibited a real correlation matrix of order sixteen and rank three, the Gram matrix of one representative from each antipodal pair of vertices of a polytope inscribed in the sphere of $\mathbb{R}^{3}$, for which $\Pi(A)$ has an eigenvalue exceeding $\operatorname{per}(A)$.

Complex rank two. Tran's matrix [9]

$$A = \begin{pmatrix} 3 & 1-2i & -1 & 1+2i & 1 \\ 1+2i & 3 & 1-2i & -1 & 1 \\ -1 & 1+2i & 3 & 1-2i & 1 \\ 1-2i & -1 & 1+2i & 3 & 1 \\ 1 & 1 & 1 & 1 & 1 \end{pmatrix}$$

is Hermitian positive semidefinite of rank two with $\operatorname{per}(A) = 504$, and 512 is an eigenvalue of $\Pi(A)$.

In the proof, complex Gram vectors $v_i \in \mathbb{C}^{2}$ would give factors $D_i = v_i v_i^{*}$, and the decomposition (9) together with Lemma 5.2 would be needed for Hermitian M. The covariance (14) is an identity of bilinear forms and extends verbatim to complex g, but it yields $\Phi(\rho_n(gg^{T})) = (\det g)^{2k}\rho_m(gg^{T})$ with the transpose, not the conjugate transpose. For real g the two coincide; for complex g they do not, and Tran's example shows that no repair is possible.

9. Computational confirmation

Theorem 3.1 was checked by exact rational arithmetic for all $0 \le k \le n/2$ and $n \le 24$, and by an independent symbolic expansion of the kernel (10) for $n \le 8$. Theorem 1.1 itself was certified by exact rational LDL^{T} elimination of $\operatorname{per}(A) I - TT^{T}$ for integer Gram vectors of orders up to seven, and the largest eigenvalue of $\Pi(A)$ was compared with $\operatorname{per}(A)$ in floating point for random

real rank-two matrices of orders up to eight, agreeing to machine precision. These computations played no role in the proof.

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